

Import Statements

```
In [1]: import os
import custom_cmap
from PIL import Image
from IPython.display import display

import matplotlib.pyplot as plt
import matplotlib as mpl
import pandas as pd

from moria import reduce_s
from pathlib import Path

from astropy.visualization import PercentileInterval
from astropy.io import fits
from astropy.visualization import LogStretch, ImageNormalize
import plotly.express as px
import numpy as np
from astropy.visualization import ImageNormalize, AsinhStretch, SqrtStretch, LogStretch, PowerStretch

plt.rcParams.update({'font.size':25})
plt.rc('text', usetex=True)
%config InlineBackend.figure_format = 'retina'
%load_ext autoreload
%autoreload 2
```

Understanding the main directory.

The data directory should be organized as follows. You can look at the sample data folder to understand the directory structure you need.

00.DATA

01.XYM

02.CMD

03.LOC_TRANS

04.PSF_EXTRACT

05.COORD_TRANS(OPTIONAL)

06.FIT

07.CALIBRATION

All the necessary scripts are included in the sample data directory. The simplest way to run MORIA on your target is to copy the entirety of the "data" folder to wherever you wish to conduct your analysis. Start by putting your exposures in "data/00.DATA", then beginning with the demo.

After you finish running this notebook, you will be able to use the "_flc" exposures to generate an output image stack in the F814W and F606W HST filters.

NOTE: `chmod +x program.src` is a useful command to use whenever permissions are denied for a script. If any of these scripts fail, each folder in the list above (00.DATA, 01.XYM, 02.CMD, etc.) has an output "log file" for the Fortran scripts we run. If the "log file" for the step you ran shows a "permission denied" error, it is very likely that you need to run `chmod _x program.src` for each ".src" file used in this pipeline.

```
In [3]: #This is the directory where you are processing your data. This does not point at MORIA.  
directory = os.getcwd()
```

Data Preparation

MORIA runs the "Makefile" in `fortran_compile` directly upon pip installation.

If you have different fortran dependencies, you will have to edit the `Makefile.txt` in `fortran_compile`, and then uncomment the `code_directory` to point to `src/fortran_compile`. Once you finish compiling the files above, run this cell below to put the fortran scripts in their appropriate directories:

```
In [4]: reduce.data_prep_early(directory)
```

STEP 1

Download the _flc files. Place them appropriately in the F814W and F606W directories in 00.DATA.

The cell below will convert the _flc files to _WJ2 files using run_convert_C1K1C.src

These "_flc" exposures include the charge-transfer efficiency (CTE) corrections. MORIA first converts these "_flc" exposures into "_WJ2" files to convert to a full-chip co-ordiante system that are placed into appropriate reference geometry for subsequent steps.

```
In [5]: reduce.run_xgf_conversion(directory)
```

```

ENTER      WFCFLTREAD: ifhl29a6q_flg.fits
WFC3UV_FLTREAD NAMEAP: UVIS-CENTER      ifhl29a6q_flg.fits
FILE0: ifhl29a6q_WJC.fits
ENTER      writfits_i2_32K: ifhl29a6q_WJC.fits

```

	64	192	320	448	576	704	832	960	1088	1216	1344	1472	1600	1728	1856	1984	2112
2240	2368	2496	2624	2752	2880	3008	3136	3264	3392	3520	3648	3776	3904	4032			
4032	5	39	10	14	12	23	8	9	63	11	53	66	21	3	16	-2	24
83	23	32	64	11	3	5	11	14	9	8	10	7	13	8			
3904	17	8	16	16	12	12	19	283	6	7	15	22	68	5	38	-1	16
9	4	8	15	24	2	31	11	12	13	24	37	6	6	24			
3776	7	9	2	7	-5	19	3	-1	2	6	30	20	78	6	19	12	15
3	14	16	17	14	2	18	5	7	4	13	9	12	22	18			
3648	35	17	12	15	12	8	11	15	20	8	12	3	16	2	9	54	17
13	-8	16	13	-1	-4	5	28	11	1	16	40	19	17	22			
3520	28	8	9	12	3	10	10	-1	16	1	7	16	12	-2	18	6	331
10	15	12	9	13	9	16	12	18	11	9	1	12	18	13			
3392	19	6	13	22	10	19	7	26	66	5	15	6	8	19	24	-3	71
16	50	5	13	9	26	10	10	20	7	2	13	17	27	23			
3264	22	5	8	-2	15	12	41	11	3	3	15	4	29	15	22	6	362
64	11	8	5	30	28	33	17	9	17	17	15	13	44	3			
3136	15	33	495	40	17	80	10	11	46	8	13	30	134	20	10	10	11
27	12	13	21	15	31	21	3	13	9	35	9	24	29	7			
3008	14	25	12	19	8	22	18	16	14	3	14	15	9	8	14	17	11
8	10	9	22	6	2	16	6	19	3	38	15	22	18	18			
2880	11	7	1	3	14	13	1	3	21	15	4	4	48	4	-10	5	73
12	34	33	5	21	13	18	18	7	15	11	2	13	11	28			
2752	8	13	13	31	43	17	9	11	5	15	14	9	31	33	35	18	10
17	18	51	27	21	10	25	83	12	15	31	28	16	41	20			
2624	-8	9	4	31	17	11	10	7	4	50	5	184	14	5	5	13	14
15	82	60	28	3	10	33	16	16	6	19	24	1	9	36			
2496	13	12	5	40	7	24	7	10	0	21	5	41	174	0	13	19	17
12	17	9	3	9	6	17	46	12	22	15	30	26	18	10			
2368	17	9	25	9	20	10	12	5	18	17	24	20	46	28	6	23	14
-1	14	29	22	11	16	13	20	8	15	27	10	49	16	5			
2240	9	16	16	56	20	11	7	-1	7	23	7	63	-2	9	12	6	20
7	14	18	11	19	19	10	-1	29	17	16	5	21	18	21			
2112	49	11	2	30	19	0	104	16	19	10	18	15	7	8	17	17	25
12	30	2	5	20	12	22	9	6	12	12	5	19	19	21			
	64	192	320	448	576	704	832	960	1088	1216	1344	1472	1600	1728	1856	1984	2112

	2240	2368	2496	2624	2752	2880	3008	3136	3264	3392	3520	3648	3776	3904	4032			
1984		8	24	6	13	26	7	-2	15	6	21	14	3	24	4	27	5	13
19	183	16	41	5	22	18	8	40	22	15	9	8	21	6				
1856		12	3	9	12	6	8	20	33	18	15	12	29	5	16	14	23	18
6	3	9	2	15	4	25	23	36	10	70	46	15	15	15				
1728		32	23	12	-1	13	7	3	32	23	5	7	16	18	8	28	18	12
28	18	10	11	6	12	36	5	17	13	940	39	26	22	3				
1600		14	66	62	6	18	12	26	42	15	2	10	5	2	3	0	7	26
6	3	18	-1	8	7	23	19	39	10	16	31	24	18	21				
1472		7	24	11	23	11	16	7	21	20	11	5	22	3	14	14	18	36
174	4	11	5	25	7	20	29	18	37	14	67	10	9	19				
1344		24	21	3	53	21	11	9	24	28	16	10	13	8	19	12	10	7
9	17	15	-3	7	-6	32	1	12	13	9	3	31	15	22				
1216		9	6	34	437	23	19	6	6	4	12	9	46	51	30	13	1	3
5	-4	11	14	9	21	10	22	19	68	21	2	27	15	8				
1088		14	17	10	12	13	5	8	8	31	163	11	17	8	8	5	7	27
10	6	12	18	27	19	17	-6	7	16	20	11	9	6	7				
960		3	15	21	15	26	6	8	18	16	22	25	12	24	20	2	-2	16
37	14	16	14	25	2	10	23	16	18	17	10	1	4	18				
832		8	21	17	10	0	663	3	18	0	5	8	13	8	8	9	-2	22
12	38	642	5	3	40	57	21	6	16	0	8	8	11	22				
704		14	6	0	7	10	8	-1	16	16	18	11	11	4	31	16	30	7
4	12	9	1	15	9	12	14	13	6	9	28	10	8	8				
576		25	20	6	5	13	19	12	15	4	12	0	20	31	10	14	7	4
84	11	-1	7	17	6	8	61	12	80	20	8	28	14	23				
448		9	15	18	0	37	28	14	15	15	3	8	18	20	10	21	24	16
9	36	58	17	60	20	0	3	5	18	20	6	10	33	17				
320		12	19	0	3	22	11	9	13	13	0	8	11	163	23	2	9	23
9	1	10	23	8	25	6	15	103	7	49	18	10	11	17				
192		81	670	1	6	20	7	9	10	13	-1	6	12	16	8	49	8	20
2	10	27	7	6	34	2	216	0	11	15	36	6	18	22				
64		2	13	56	10	14	6	9	21	14	108	15	4	18	8	5	10	22
8	38	13	8	10	7	13	6	18	22	9	12	30	8	14				
		64	192	320	448	576	704	832	960	1088	1216	1344	1472	1600	1728	1856	1984	2112
2240	2368	2496	2624	2752	2880	3008	3136	3264	3392	3520	3648	3776	3904	4032				

ENTER WFCFLTREAD: ifhl29a8q_flg.fits
WFC3UV_FLTREAD NAMEAP: UVIS-CENTER ifhl29a8q_flg.fits
FILE0: ifhl29a8q_WJC.fits

ENTER writfits_i2_32K: ifhl29a8q_WJC.fits

	64	192	320	448	576	704	832	960	1088	1216	1344	1472	1600	1728	1856	1984	2112
2240	2368	2496	2624	2752	2880	3008	3136	3264	3392	3520	3648	3776	3904	4032			
4032	91	77	32	86	27	21	54	43	27	45	58	199	52	27	24	17	27
26	35	106	121	42	31	55	44	23	39	39	32	2435	41	43			
3904	18	39	28	41	37	35	57	44	29	127	44	48	129	36	14	51	16
151	35	27	41	13	15	39	36	26	53	30	30	238	32	26			
3776	151	144	32	51	59	39	48	53	368	49	35	163	66	78	30	35	51
43	44	121	29	21	37	58	23	29	122	148	32	58	83	105			
3648	36	39	108	25	40	55	89	32	269	309	28	79	52	59	49	24	21
68	157	60	45	37	39	33	24	149	23	29	23	100	67	48			
3520	607	76	206	45	62	57	23	47	30	32	49	88	40	72	104	1923	42
46	46	31	41	30	15	25	28	34	16	31	131	36	34	43			
3392	108	43	47	22	18	29	35	30	23	46	18	65	36	44	37	46	52
32	61	40	29	65	80	34	36	20	19	32	390	40	66	31			
3264	63	42	65	47	39	54	57	35	41	64	33	44	47	38	219	37	64
30	56	44	68	38	18	28	209	35	19	73	48	57	46	27			
3136	31	27	246	57641	34	32	26	37	60	27	30	47	60	36	39	80	35
54	34	32	113	28	66	33	186	41	26	39	25	56	36	22			
3008	16	37	22	25	34	31	37	94	22	55	109	27	30	65	1286	41	39
44	35	303	34	27	50	36	42	23	21	47	73	88	34	21			
2880	34	53	42	37	54	41	29	49	21	43	240	22	64	45	39	25	50
53	52	94	55	30	47	78	54	67	34	44	33	34	29	38			
2752	40	77	34	59	93	40	22	23	29	39	62	22	41	33	33	33	46
46	48	40	823	29	50	36	55	29	32	108	17	424	241	27			
2624	28	403	28	48	29	37	58	38	30	28	42	42	28	18	32	28	35
74	39	36	162	32	152	59	90	43	48	39	33	76	114	36			
2496	34	123	29	38	38	65	34	26	97	27	85	46	92	42	29	52	47
47	67	24	1	38	20	35	27	46	23	36	57	31	67	7574			
2368	49	57	42	157	18	55	79	30	35	45	26	37	74	56	39	26	74
53	62	46	118	33	61	277	30	28	24	55	33	56	28	20			
2240	28	39	31	80	42	30	30	53	38	44	149	46	269	158	52	33	11
41	40	39	34	41	36	46	23	46	31	30	30	25	241	35			
2112	35	29	38	45	65	34	31	23	178	31	55	38	35	242	252	60	28
173	35	42	78	70	46	29	37	18	47	1700	82	112	35	37			
	64	192	320	448	576	704	832	960	1088	1216	1344	1472	1600	1728	1856	1984	2112
2240	2368	2496	2624	2752	2880	3008	3136	3264	3392	3520	3648	3776	3904	4032			
1984	27	35	42	71	24	42	43	32	44	47	19	23	18	373	64	70	65

```
ENTER      WFCFLTREAD: ifhl29acq_flg.fits
WFC3UV_FLTREAD NAMEAP: UVIS-CENTER          ifhl29acq_flg.fits
FILE0: ifhl29acq_WJC.fits
ENTER      writfits_i2_32K: ifhl29acq_WJC.fits
```

64 192 320 448 576 704 832 960 1088 1216 1344 1472 1600 1728 1856 1984 2112

	2240	2368	2496	2624	2752	2880	3008	3136	3264	3392	3520	3648	3776	3904	4032			
4032		59	83	35	37	42	20	54	39	30	40	50	52	57	36	35	43	52
43	55	64	63	28	43	61	40	37	48	71	27	1814	43	35				
3904	39	52	53	47	30	42	76	62	55	41	31	66	52	22	39	50	42	
66	46	34	33	30	46	52	54	40	39	58	38	50	37	45				
3776	153	108	44	53	32	51	68	48	72	53	76	375	45	49	54	49	46	
48	57	58	26	37	25	67	37	37	77	82	55	45	74	40				
3648	40	43	50	58	49	54	54	44	407	44	56	74	84	66	75	34	31	
55	449	217	70	33	37	49	44	48	38	66	46	52	77	48				
3520	48	65	160	72	56	48	51	50	45	51	38	74	50	64	158	77	48	
45	55	33	47	54	49	43	76	47	40	57	78	47	52	41				
3392	43	129	62	24	53	31	53	44	43	37	32	46	44	47	40	50	48	
45	87	58	43	40	101	36	32	61	51	45	61	41	65	41				
3264	99	60	61	42	47	35	45	53	46	28	55	34	52	56	55	49	50	
46	53	54	60	33	29	43	53	41	54	27	34	102	39	23				
3136	40	45	99	1997	44	41	50	37	48	29	36	47	347	53	52	64	41	
43	64	51	56	50	193	38	104	58	44	42	40	76	37	44				
3008	28	38	27	45	45	43	33	59	34	53	54	43	38	58	76	44	54	
32	47	58	50	43	57	37	61	49	48	59	41	82	27	35				
2880	58	43	57	33	45	77	37	51	42	44	74	73	126	40	34	41	87	
45	53	75	43	68	71	92	53	268	48	43	50	41	45	51				
2752	32	69	31	59	322	47	47	49	43	46	55	37	46	50	50	51	51	
42	36	44	277	58	47	58	51	45	50	50	224	61	143	46				
2624	43	64	47	43	44	53	51	38	50	37	29	44	34	40	47	44	50	
47	53	39	54	30	73	59	77	28	24	34	48	126	69	36				
2496	52	54	44	21	53	39	43	37	42	55	37	68	45	25	36	58	48	
40	46	42	27	50	53	36	47	46	27	42	66	37	49	314				
2368	30	65	42	32	40	72	42	66	29	58	32	33	25	48	65	41	55	
52	38	41	55	50	69	57	35	35	51	41	38	52	44	38				
2240	41	49	61	51	93	40	54	48	51	38	63	28	113	154	87	56	50	
44	72	37	29	67	45	46	47	54	42	44	45	49	63	38				
2112	37	43	40	30	53	42	69	40	55	36	33	49	44	70	50	50	49	
65	43	44	50	114	42	37	40	43	51	331	42	51	68	47				
	64	192	320	448	576	704	832	960	1088	1216	1344	1472	1600	1728	1856	1984	2112	
	2240	2368	2496	2624	2752	2880	3008	3136	3264	3392	3520	3648	3776	3904	4032			
1984	42	62	608	47	47	43	40	30	42	55	48	35	28	52	57	73	56	
33	51	240	35	32	45	54	41	53	106	862	80	35	57	40				
1856	55	38	36	41	40	39	66	45	47	37	59	31	64	38	44	57	56	
47	58	66	40	46	85	43	56	39	35	66	57	56	70	57				


```

1728    64    66    37    45    34    37    42    39    40    49    39    49    48    58    42    51    92
61    52    49    54    98    68    204    46    33    52    72    47    41    64    44
1600    46    37    41    168    50    37    41    34    61    56    35    49    62    43    23    60    63
31    54    39    61    83    48    55    38    41    72    47    55    34    44    50
1472    61    52    63    57    78    43    35    57    43    48    63    40    112    46    48    264    45
48    49    37    55    58    36    39    51    61    65    45    48    42    43    54
1344    40    52    21    144    69    56    45    37    40    34    45    34    32    58    40    60    44
54    41    61    96    56    39    48    65    59    34    40    48    56    43    51
1216    57    31    55    104    52    99    34    33    20    37624    46    280    36    44    36    37    100
37    44    46    53    80    57    59    35    56    54    47    42    50    74    49
1088    60    26    50    37    41    51    87    56    36    29    40    54    49    48    44    34    114
32    55    48    47    45    43    41    50    58    34    34    57    78    35    75
960    46    60    42    45    41    52    41    68    313    61    56    59    57    65    56    36    88
35    36    39    94    35    37    31    54    44    42    41    49    52    44    44
832    64    738    57    43    49    44    33    52    61    42    49    44    59    73    44    201    88
33    51    28    46    46    42    45    40    40    53    47    51    41    55    38
704    40    32    43    42    43    46    66    33    43    185    41    34    40    41    36    56    48
53    40    53    33    56    38    49    44    55    33    40    47    42    112    60
576    34    57    39    36    53    57    175    52    899    856    123    68    44    57    57    44    56
46    55    56    44    38    37    39    77    184    117    44    61    42    46    46
448    42    48    42    57    52    61    56    38    119    43    42    36    40    32    52    66    289
27    58    72    54    51    37    45    51    59    61    45    47    56    31    39
320    117    55    65    41    60    49    40    42    39    84    30    43    39    37    51    29    112
40    53    51    39    32    46    60    51    41    36    55    1625    53    52    61
192    64    82    63    44    61    81    24    61    43    47    29    50    42    43    42    42    40
45    49    107    57    44    35    39    83    63    48    29    78    42    44    48
64    74    592    305    57    43    55    48    45    89    40    44    29    131    54    34    25    36
44    52    43    26    39    60    47    40    53    47    82    68    150    47    69

```

```

        64    192    320    448    576    704    832    960    1088    1216    1344    1472    1600    1728    1856    1984    2112
2240  2368  2496  2624  2752  2880  3008  3136  3264  3392  3520  3648  3776  3904  4032

```

```

ENTER      WFCFLTREAD: ifhl29agq_flg.fits
WFC3UV_FLTREAD NAMEAP: UVIS-CENTER      ifhl29agq_flg.fits
FILE0: ifhl29agq_WJC.fits
ENTER      writfits_i2_32K: ifhl29agq_WJC.fits

```

```

        64    192    320    448    576    704    832    960    1088    1216    1344    1472    1600    1728    1856    1984    2112
2240  2368  2496  2624  2752  2880  3008  3136  3264  3392  3520  3648  3776  3904  4032

```

```

4032    36    32    43    32    37    31    28    39    83    81    37    45    29    40    35    35    38

```

53	24	69	73	32	34	33	35	27	50	50	30	45	49	35				
3904		43	27	27	42	47	28	28	46	36	37	44	50	176	36	51	38	140
102	24	26	60	84	29	41	31	22	41	38	34	34	31	37				
3776		24	21	38	24	50	30	33	33	34	37	101	39	47	16	24	32	48
34	37	48	28	29	38	54	29	38	29	32	57	31	51	39				
3648		37	31	22	42	53	41	43	26	55	29	34	39	34	36	35	73	28
30	27	24	26	26	27	44	40	33	26	37	39	28	58	47				
3520	120	27	31	33	35	43	36	18	66	35	25	24	45	43	37	34	119	
58	34	18	37	38	36	113	151	30	42	35	29	138	45	87				
3392		94	40	38	28	46	44	50	49	41	49	31	33	37	27	28	39	121
39	32	32	35	34	33	63	46	36	34	25	37	37	41	77				
3264		65	25	37	29	40	38	22	40	22	38	37	41	42	41	53	42	3909
61	33	40	43	36	51	58	36	35	661	91	41	42	63	34				
3136		42	34	694	42	38	46	29	42	33	39	40	34	112	32	32	53	57
43	41	51	38	57	42	31	52	38	39	53	52	40	66	46				
3008	123		34	19	36	26	309	25	36	40	76	34	36	28	39	105	57	47
50	51	36	56	44	88	43	28	48	32	29	35	91	50	19				
2880		38	43	36	40	34	56	28	29	27	32	43	34	104	44	40	36	67
36	49	65	35	39	43	62	58	29	55	39	36	86	45	36				
2752		28	53	60	31	73	32	38	59	97	28	34	38	32	55	44	43	38
79	38	46	41	41	31	34	31	44	48	40	40	28	88	38				
2624		29	33	53	38	58	43	29	33	22	33	69	40	34	34	25	44	62
425	38	33	39	44	34	46	42	51	54	53	43	36	34	49				
2496		29	35	54	36	37	80	36	29	35	49	33	46	21	21	22	86	32
206	48	37	33	37	64	45	32	39	56	25	31	35	35	64				
2368		48	43	42	26	42	35	30	223	33	23	38	51	48	48	43	41	41
254	31	42	28	42	40	30	50	34	31	40	25	266	58	36				
2240		26	50	37	47	44	41	22	39	40	50	33	74	24	36	24	34	22
49	43	38	37	36	58	30	31	36	37	26	880	67	32	45				
2112		97	40	53	48	26	29	51	14	41	48	70	43	25	40	32	46	104
48	26	31	31	68	45	40	42	36	28	34	34	40	29	55				
		64	192	320	448	576	704	832	960	1088	1216	1344	1472	1600	1728	1856	1984	2112
2240	2368	2496	2624	2752	2880	3008	3136	3264	3392	3520	3648	3776	3904	4032				
1984		44	42	30	43	38	39	28	38	51	67	52	37	128	54	30	39	33
53	45	42	35	43	34	31	39	49	29	49	47	153	52	37				
1856		40	35	54	52	47	29	30	66	87	40	44	31	41	31	45	73	39
49	37	39	51	48	41	35	51	71	32	64	34	49	26	31				
1728		46	27	47	36	152	34	27	189	68	49	47	33	49	36	99	38	35
42	34	51	33	56	39	31	21	61	33	172	40	137	31	21				
1600		34	32	148	38	50	29	92	35	35	33	33	42	25	34	25	46	41

33	41	38	49	36	33	49	43	34	47	27	40	428	53	45				
1472		36	30	45	49	86	82	31	41	26	32	62	46	40	63	34	39	51
39	35	28	27	27	42	44	56	33	107	41	80	30	52	43				
1344		37	36	61	142	130	60	30	35	47	38	29	76	50	38	23	139	38
30	78	37	37	28	35	123	29	28	34	46	13	46	44	32				
1216		37	35	55	1387	69	39	42	20	51	40	64	48	32	50	40	21	27
36	59	170	36	33	27	30	38	51	77	44	48	46	74	24				
1088		38	42	35	37	38	53	37	29	45	36	28	28	47	33	31	30	32
34	28	24	39	27	49	132	32	36	51	14	33	38	43	42				
960		39	62	49	31	19	43	34	47	39	26	38	48	19	42	26	36	41
230	30	32	30	50	31	33	206	39	39	40	12	25	31	40				
832		30	33	43	37	46	44	31	43	40	39	33	47	25	32	49	31	61
43	61	107	35	38	37	30	35	32	61	26	46	19	19	35				
704		33	28	28	36	28	26	18	19	46	42	28	34	53	46	38	34	28
38	70	71	29	42	49	38	10	39	23	32	45	63	60	46				
576		29	29	35	44	21	34	43	36	48	43	32	40	48	31	29	307	38
47	53	41	38	36	45	75	47	44	88	36	32	46	31	52				
448		49	26	37	41	31	775	27	41	23	30	37	32	30	34	48	45	45
40	44	25	52	56	81	33	30	36	36	46	37	44	31	66				
320		44	26	35	31	22	26	34	36	45	42	32	26	54	23	28	34	49
28	38	44	56	30	32	32	37	367	81	26	29	43	39	31				
192		33	50	43	19	43	31	15	29	36	22	27	31	45	39	45	37	46
30	33	23	32	44	29	42	62	29	44	36	38	47	29	37				
64		27	34	32	32	29	33	38	45	39	34	32	43	30	30	37	24	38
33	40	35	49	37	20	33	34	40	32	26	62	42	42	28				
		64	192	320	448	576	704	832	960	1088	1216	1344	1472	1600	1728	1856	1984	2112
2240	2368	2496	2624	2752	2880	3008	3136	3264	3392	3520	3648	3776	3904	4032				

Step 2

Prepare the IN.* files taken in by the .src scripts in 01.XYM and 02.CMD directories. This step is important since the IN.* files in each directory are used to provide input to the backend Fortran scripts.

```
In [6]: reduce.data_prep(directory)
```

Creature MATCHUP files

In the cell below we will achieve the following outcomes:

We will use respective PSF files for each filter to create a .XYM file for each exposure. A library PSF model is used to produce ".XYM" files, containing the star positions and magnitudes for each HST image exposure.

Generate output files TRANS.xym2mat, along with 16 MAT.0 files. Note that the first line of the IN.xym2mat and files beginning with 00 just defines the reference coordinates to be used for the output, so the 00 file does not need to be a *_WJ2.xym file.

Find all the objects that could be found in at least 12 out of 16 exposures.

```
In [7]: reduce.matchup_files(directory)
```

Let's open the MATCHUP file to inspect it further. The MATCHUP file is stored in 01.XYM/F814W for the F814W filter and 01.XYM/F606W for the F606W filter

The MATCHUP file we will open will be for the F814W folder. You can uncomment the filename_606W line in the cell below to look at the F606W MATCHUP file.

```
In [8]: filename = Path(directory).resolve()/f"01.XYM/dex_no_gaia_STEP08_A.xyviiiiii"
cols = ["xbar", "ybar", "mv", "mi"]
df = pd.read_csv(filename, sep=r"\s+", comment="#", usecols = [0,1,2,3], header=None, names=cols)
```

```
In [9]: df
```

Out [9]:

	xbar	ybar	mv	mi
0	1784.674	3601.858	-7.088	-8.537
1	1792.810	3612.922	-8.228	-9.719
2	1804.102	3602.212	-7.512	-9.108
3	1811.062	3623.046	-11.393	-11.941
4	1815.353	3606.928	-6.911	-8.493
...	...	...	...	...
62833	8189.384	6292.003	-7.669	-9.266
62834	8193.288	6314.011	-8.291	-10.038
62835	8193.363	6352.835	-7.738	-9.099
62836	8209.630	6308.120	-7.380	-8.993
62837	8227.282	6341.417	-10.711	-12.804

62838 rows x 4 columns

From the MATCHUP file above, here are the most important things to note:

1. xbar and ybar correspond to the x and y star positions (in pixel coordinates) for stars in the HST images.
2. mv is the F606W V magnitude of the stars.
3. mi is the F814W I magnitude of the stars. Using the MATCHUP files, we have also created an HST output image stack in the F606W and F814W filters. Let's look at them.

You can view the output stack in the jupyter notebook [here](#). This is a down-scaled version because the .fits file is too large to open in the notebook directly. The output stacks for both filters are in 01.XYM.

We open the output stack for F814W below. If you want to see the upscaled true version of the FITS file, use ds9 on your terminal.

The scaling can be adjusted depending on your preference by changing the argument given to "ImageNormalize" in the cell below.

```

In [10]: filename_814W = Path(directory).resolve() / "01.XYM/dex_no_gaia_STEP10B_F814W.fits"
filename_606W = Path(directory).resolve() / "01.XYM/dex_no_gaia_STEP10A_F606W.fits"

# Full stacks are often 10k+ pixels; loading + Plotly serializes the entire array and kills the kernel.
# Keep only a strided preview in RAM for visualization (full FITS on disk is unchanged).
MAX_PREVIEW_SIDE = 1536

def read_stack_preview(path, max_side=MAX_PREVIEW_SIDE):
    with fits.open(path, memmap=True) as hdul:
        raw = hdul[-1].data
        ny, nx = int(raw.shape[0]), int(raw.shape[1])
        step = max(int(np.ceil(max(ny, nx) / max_side)), 1)
        # Small contiguous copy; stride read from memmap avoids holding the full image in memory.
        preview = np.array(raw[:,::step, ::step], dtype=np.float32, copy=True)
        preview = np.nan_to_num(preview, nan=0.0, posinf=0.0, neginf=0.0)
        return preview, step, (ny, nx)

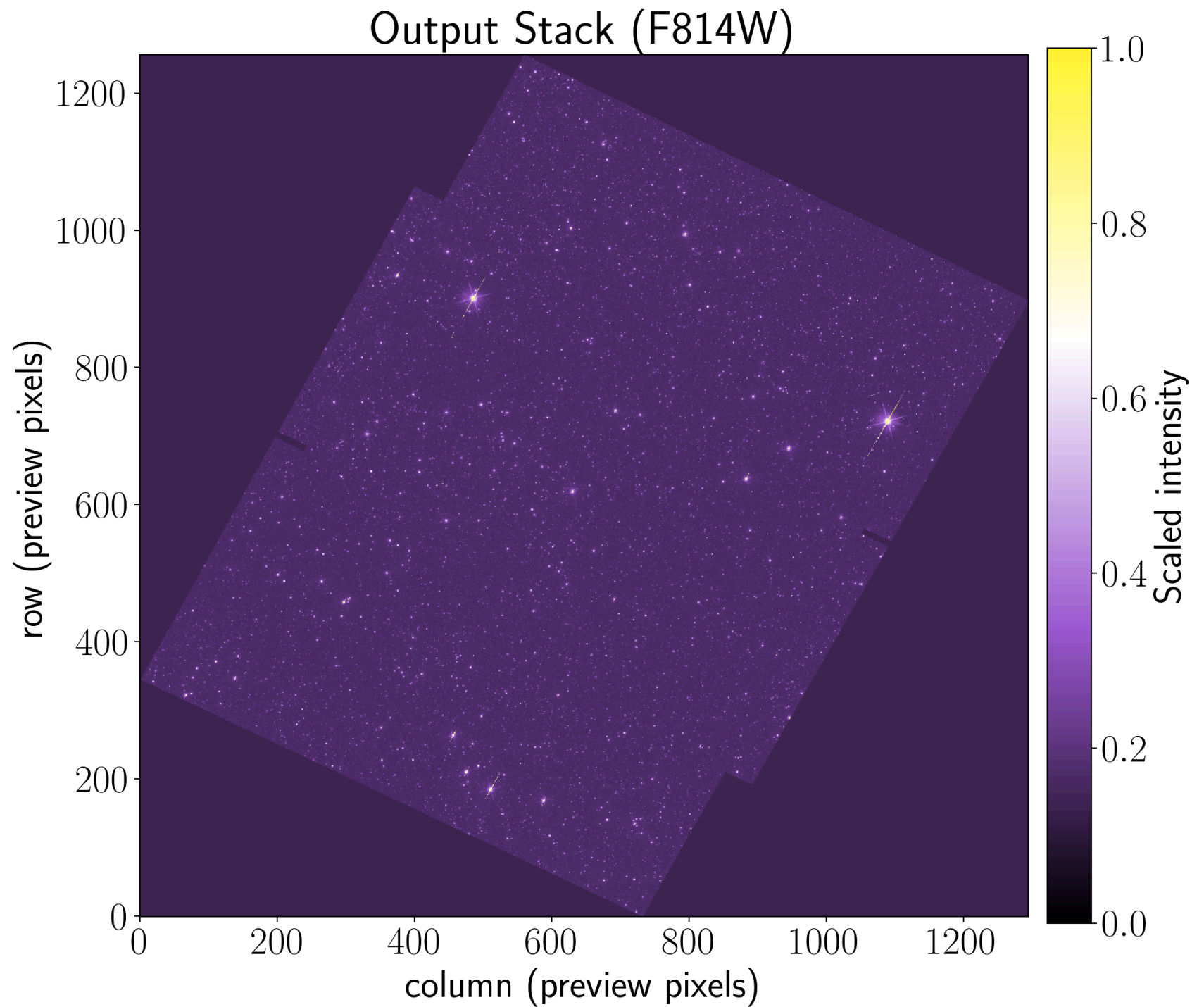
preview, stride, full_shape = read_stack_preview(filename_814W)
norm = ImageNormalize(preview, stretch=AsinhStretch(0.001))
scaled = norm(preview)

cmap = custom_cmap.mpl_nb()
fig, ax = plt.subplots(figsize=(11, 11))
im = ax.imshow(scaled, origin="lower", cmap=cmap, interpolation="nearest")
ax.set_title(f"Output Stack (F814W)")
ax.set_xlabel("column (preview pixels)")
ax.set_ylabel("row (preview pixels)")
plt.colorbar(im, ax=ax, fraction=0.046, pad=0.02, label="Scaled intensity")
plt.show()

```

WARNING: non-ASCII characters are present in the FITS file header and have been replaced by "?" characters [astropy.io.fits.util]

WARNING: Header block contains null bytes instead of spaces for padding, and is not FITS-compliant. Nulls may be replaced with spaces upon writing. [astropy.io.fits.header]



In []: